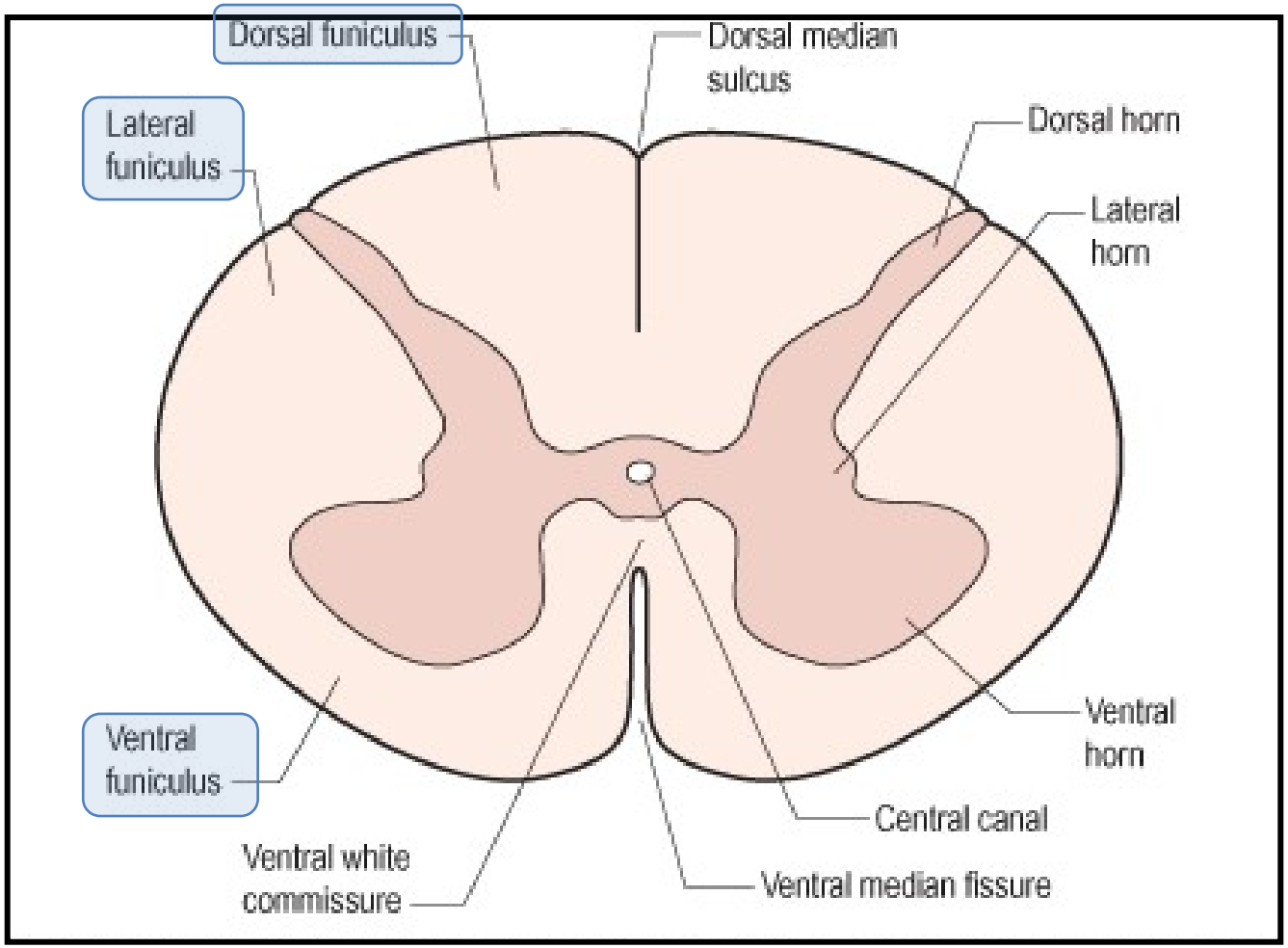
# ASCENDING SPINAL TRACTS

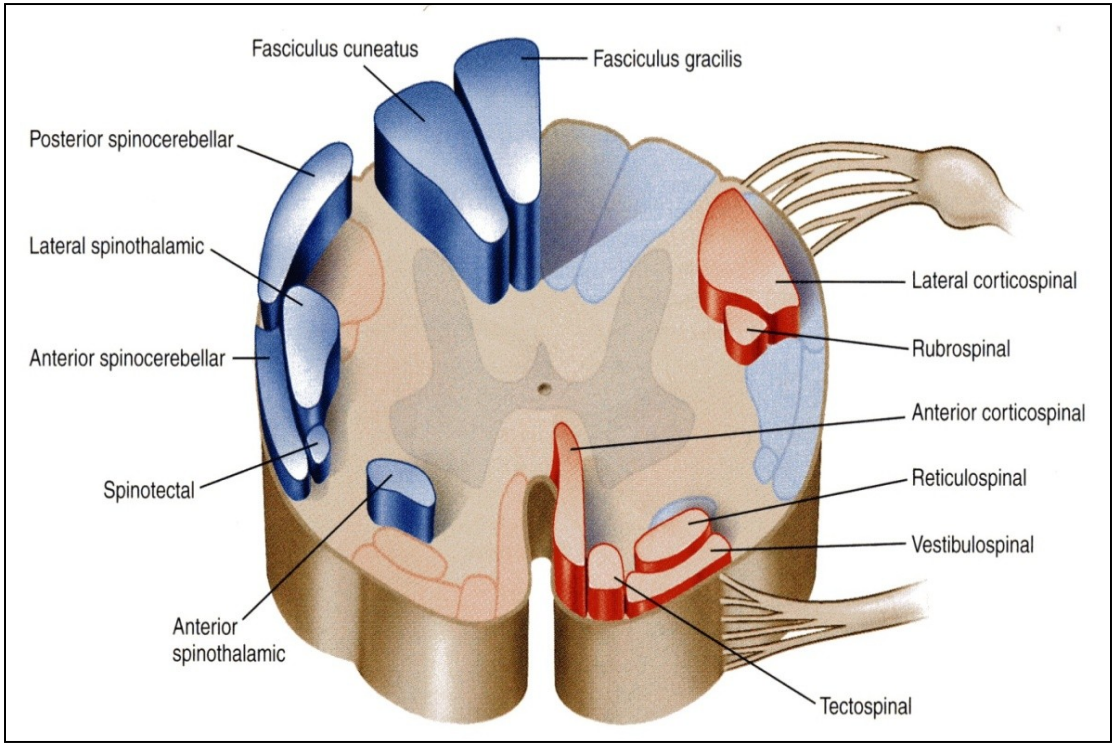





- The **White matter** of the spinal cord consists of
- **Ascending** and **Descending** **Nerve Fibers.**
- It is divided into ***Dorsal, Lateral*** & ***Ventral Columns*** or **Funiculi**.

# WHITE MATTER TRACTS

- *Bundles or fasciculi of fibers that occupy more or less definite positions in the white matter.*
- *They have the same* ***Origin***, ***Termination*** *and carry the same* ***Function.***



# WHITE MATTER TRACTS

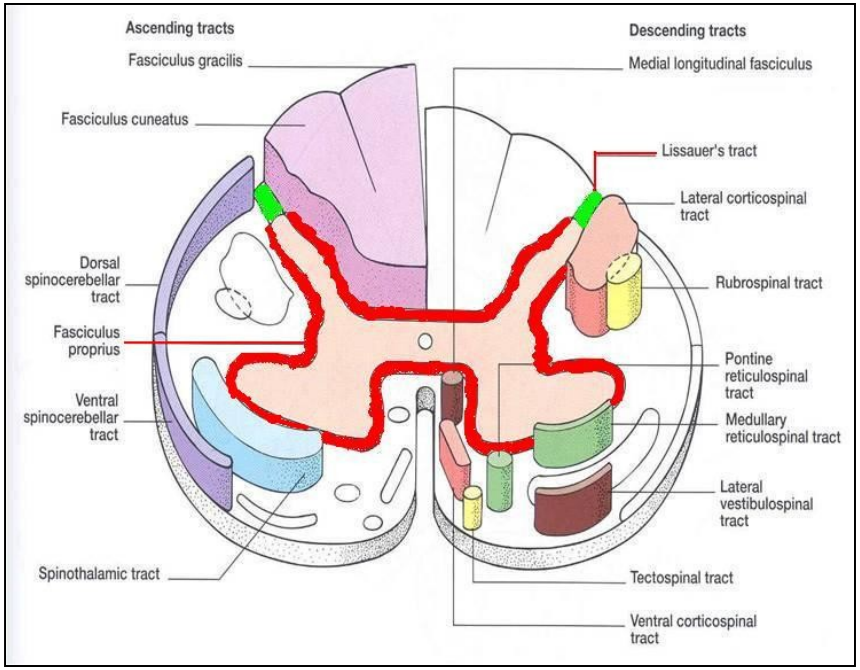
- ***They are classified into*:**

  1- ***Short Tracts;***
  ***intersegmental or propriospinal).***

*Fibers occupy narrow band peripheral to the grey matter (**fasciculus proprius**)*

*They interconnect adjacent or distant spinal segments*

*And Permit **intersegmental coordination***

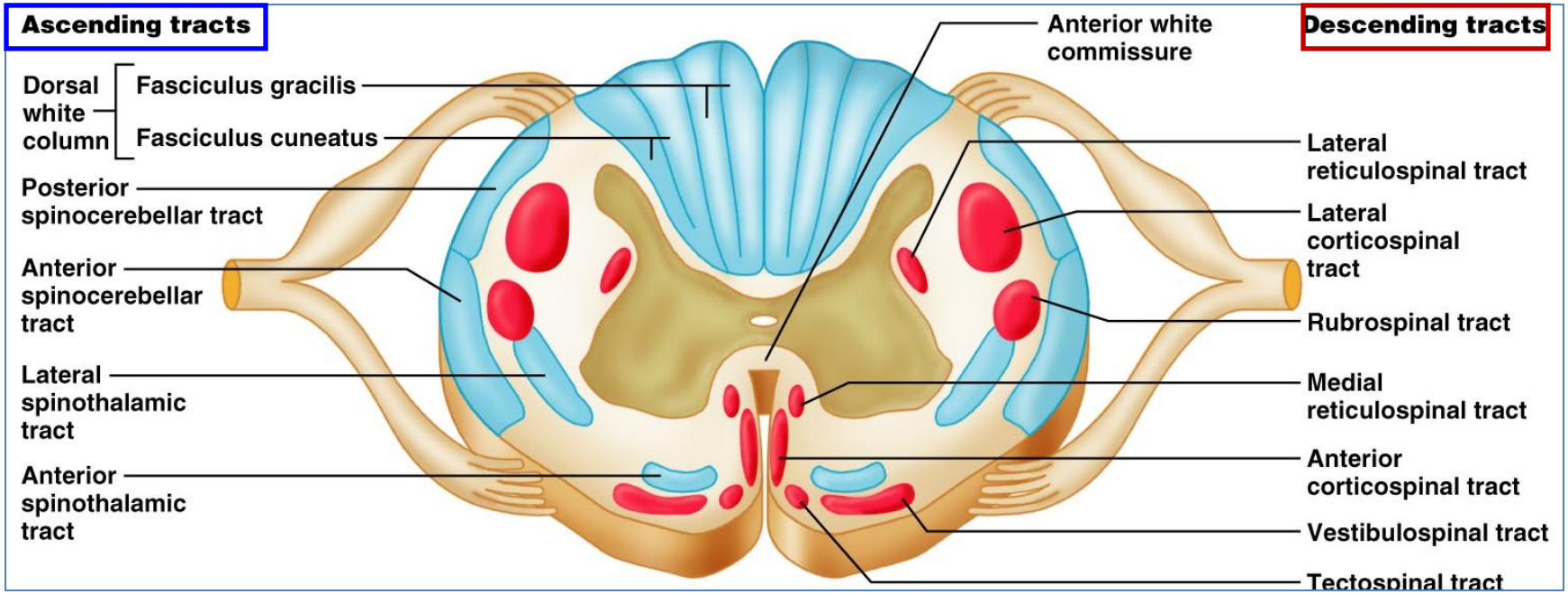


## 2-Long Tracts:

### (a) Ascending (sensory or afferent).

### (b) Descending (motor or efferent).

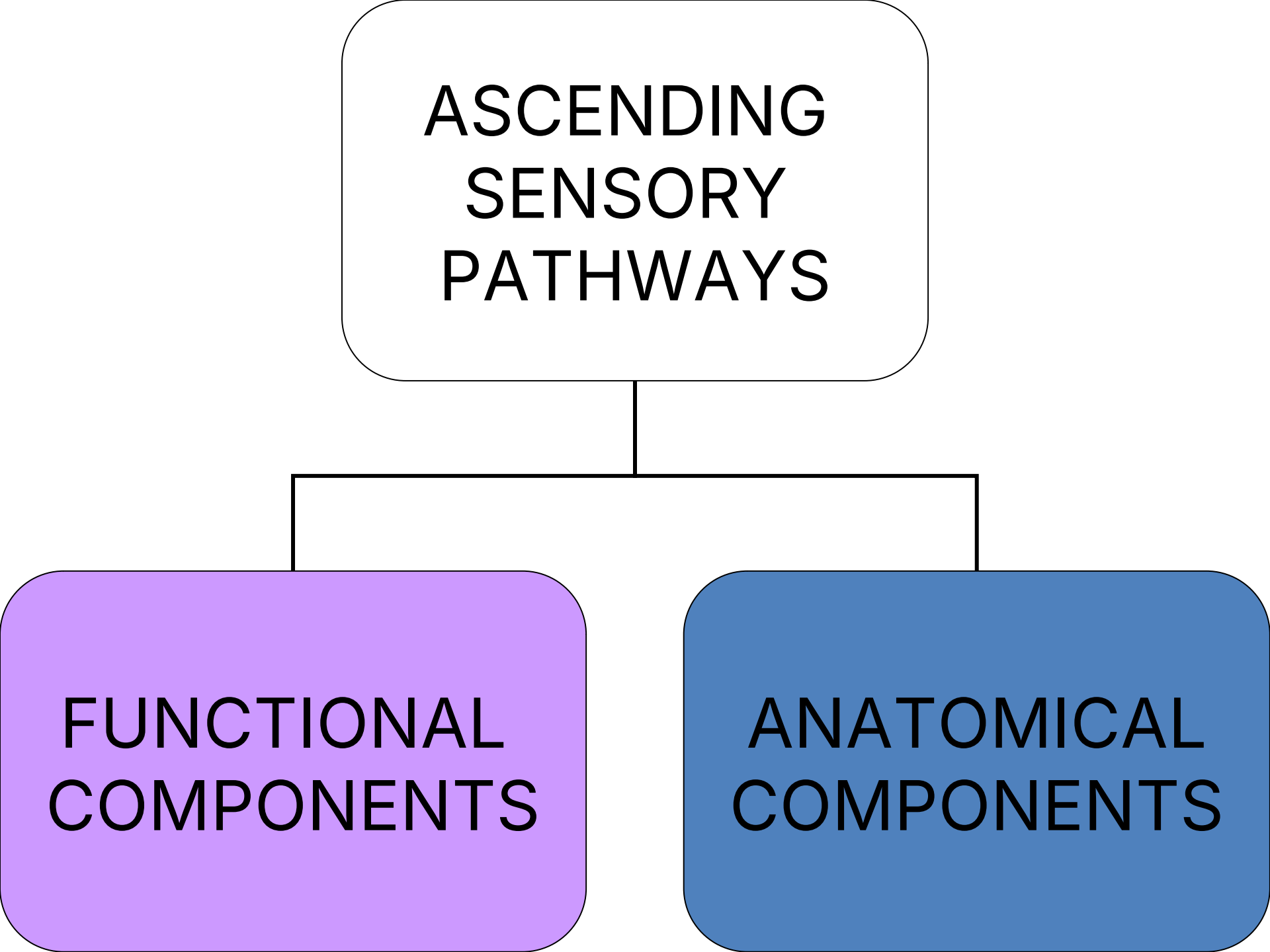
*They* serve to join the brain to the spinal cord.



# Sensory and Motor Tracts

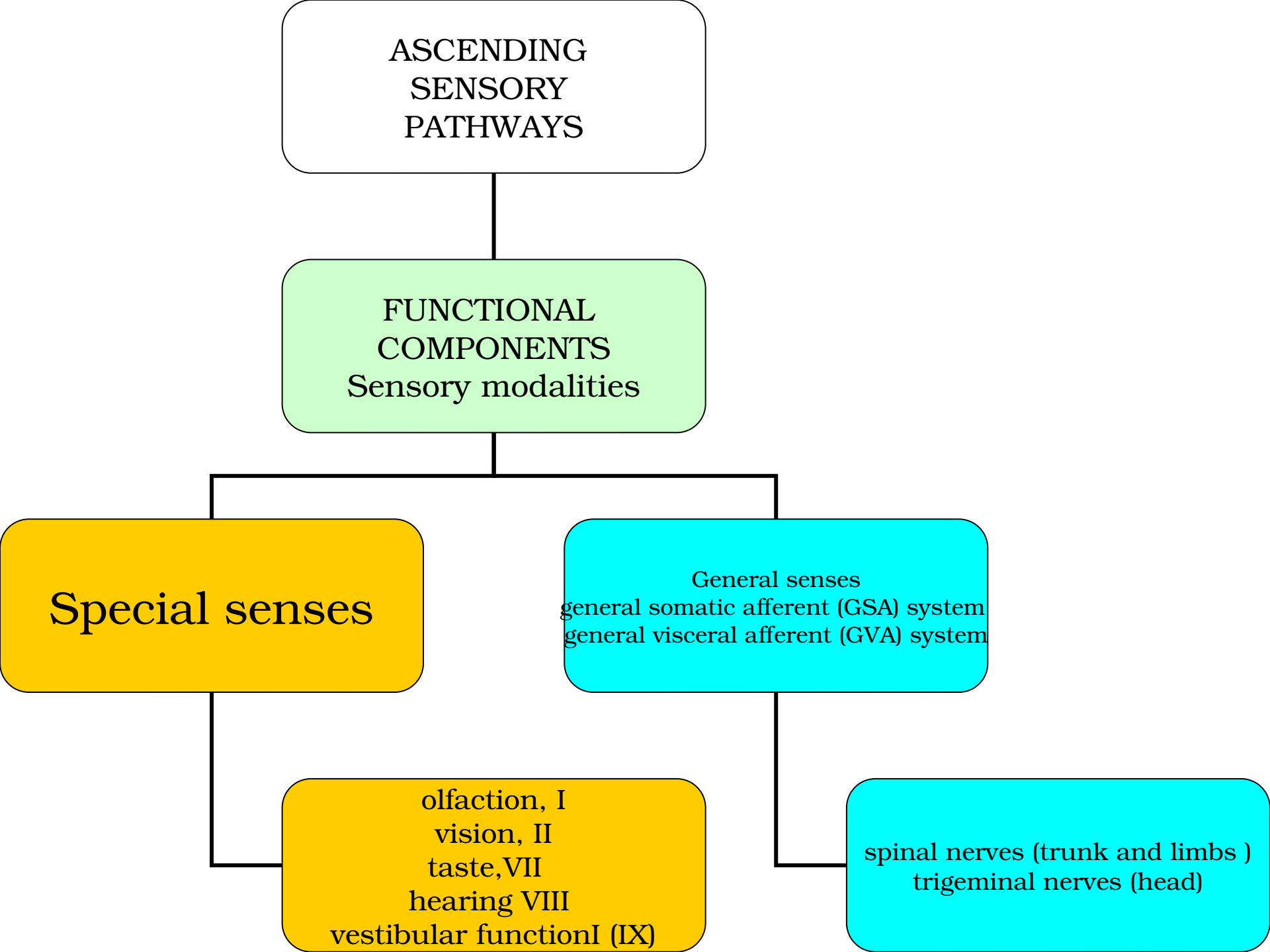
Naming the tracts

- If the tract name begins with “spino” (as in spinocerebellar), the tract is a sensory tract delivering information from the spinal cord to the cerebellum (in this case)
- If the tract name ends with “spinal” (as in vestibulospinal), the tract is a motor tract that delivers information from the vestibular apparatus (in this case) to the spinal cord


ASCENDING SENSORY PATHWAYS
FUNCTIONAL COMPONENTS
ANATOMICAL COMPONENTS

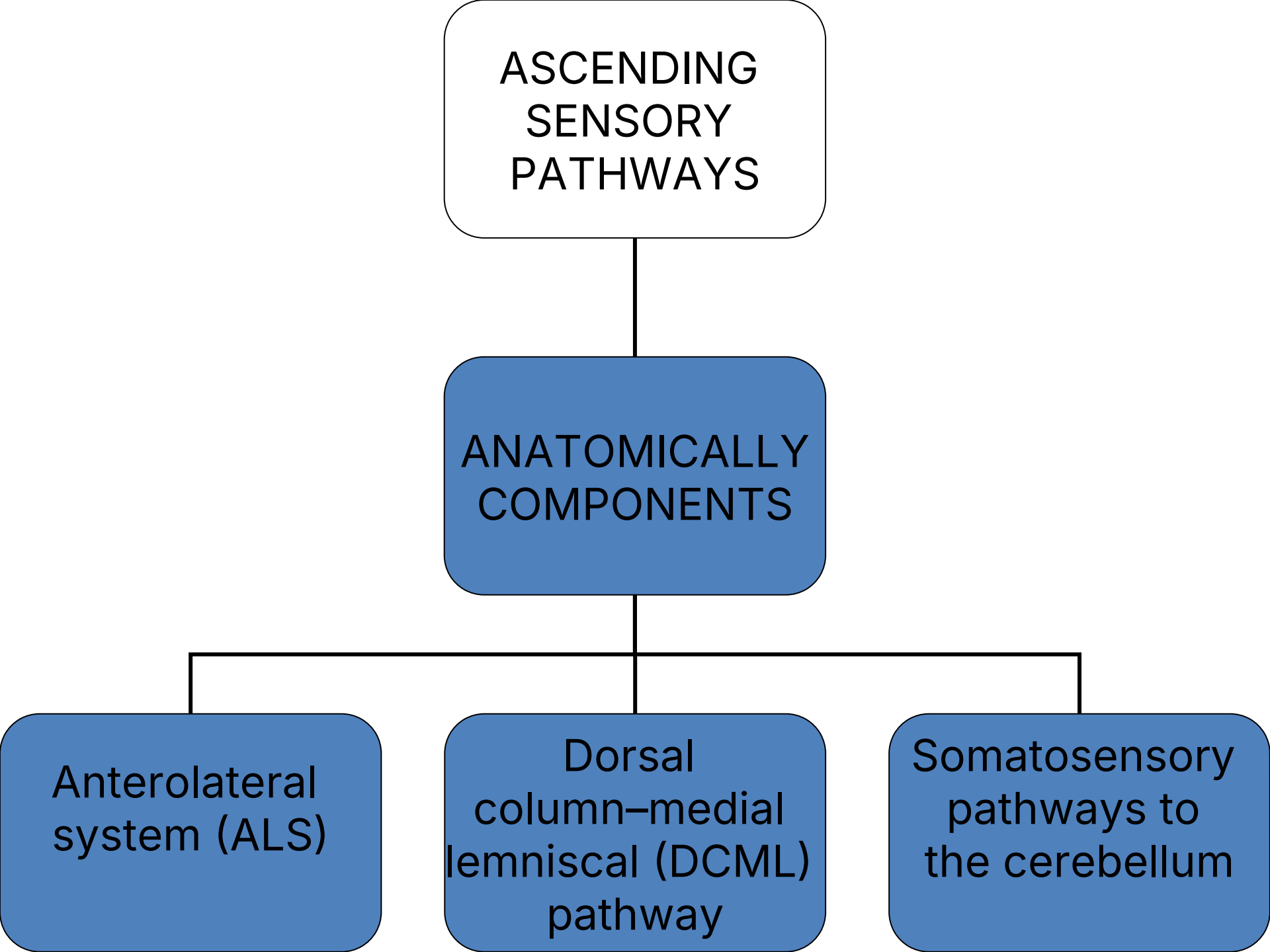
# Ascending sensory pathways

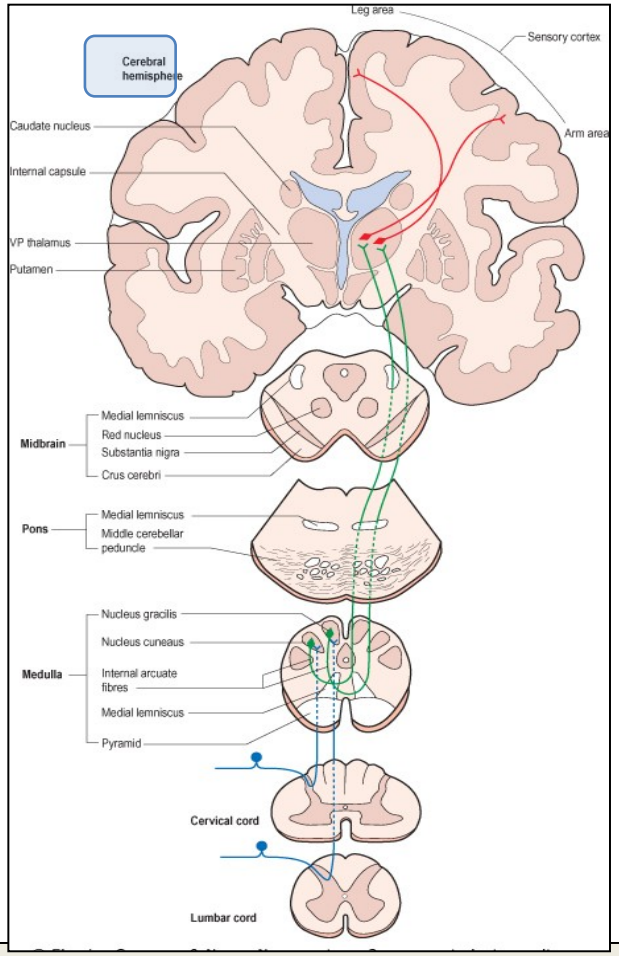
CLASSIFICATION ACCORDING TO FUNCTIONAL COMPONENTS

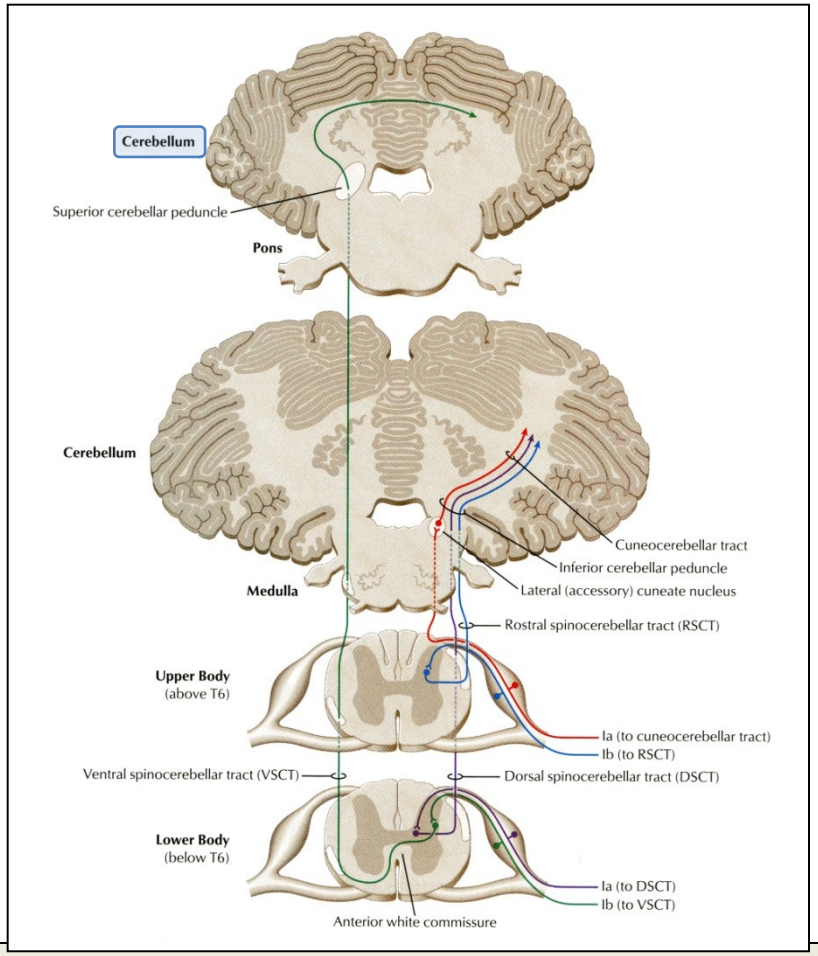

ASCENDING
SENSORY
PATHWAYS
FUNCTIONAL
COMPONENTS
Sensory modalities
Special senses
General senses
general somatic afferent (GSA) system
general visceral afferent (GVA) system
olfaction, I
vision, II
taste,VII
hearing VIII
vestibular functionI (IX)
spinal nerves (trunk and limbs )
trigeminal nerves (head)

# Ascending sensory pathways

CLASSIFICATION ACCORDING TO ANATOMICAL components

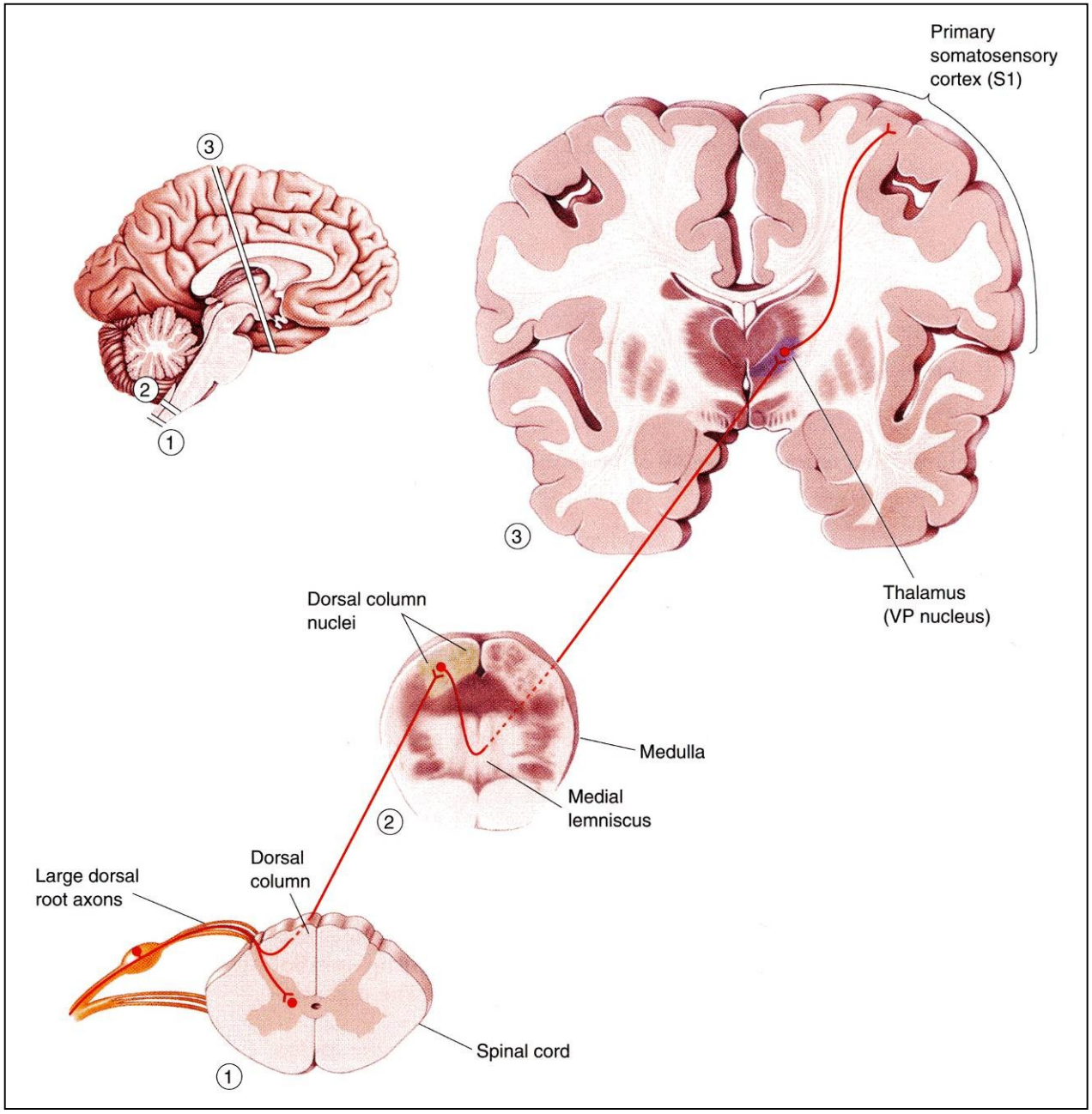

ASCENDING SENSORY PATHWAYS
ANATOMICALLY COMPONENTS
Anterolateral system (ALS)
Dorsal column–medial lemniscal (DCML) pathway
Somatosensory pathways to the cerebellum




## ❑ Ascending Tracts;

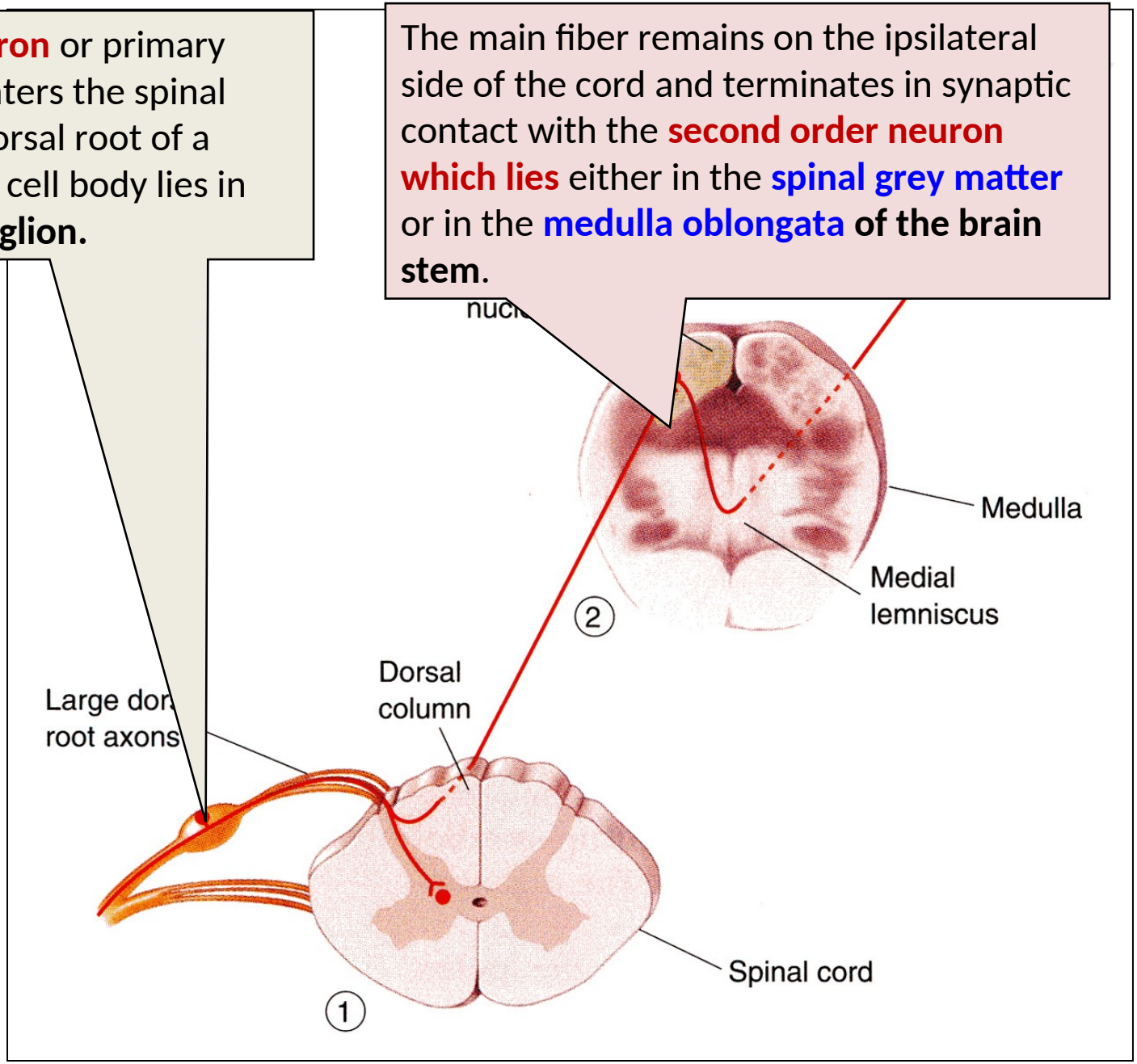
- Carry impulses from pain, thermal, tactile, muscle and joint receptors to the brain.
- *Some of this information eventually reaches a* ***conscious level*** (the **cerebral cortex),**
- *while some is destined for* ***subconscious centers*** (e.g. the **cerebellum**).



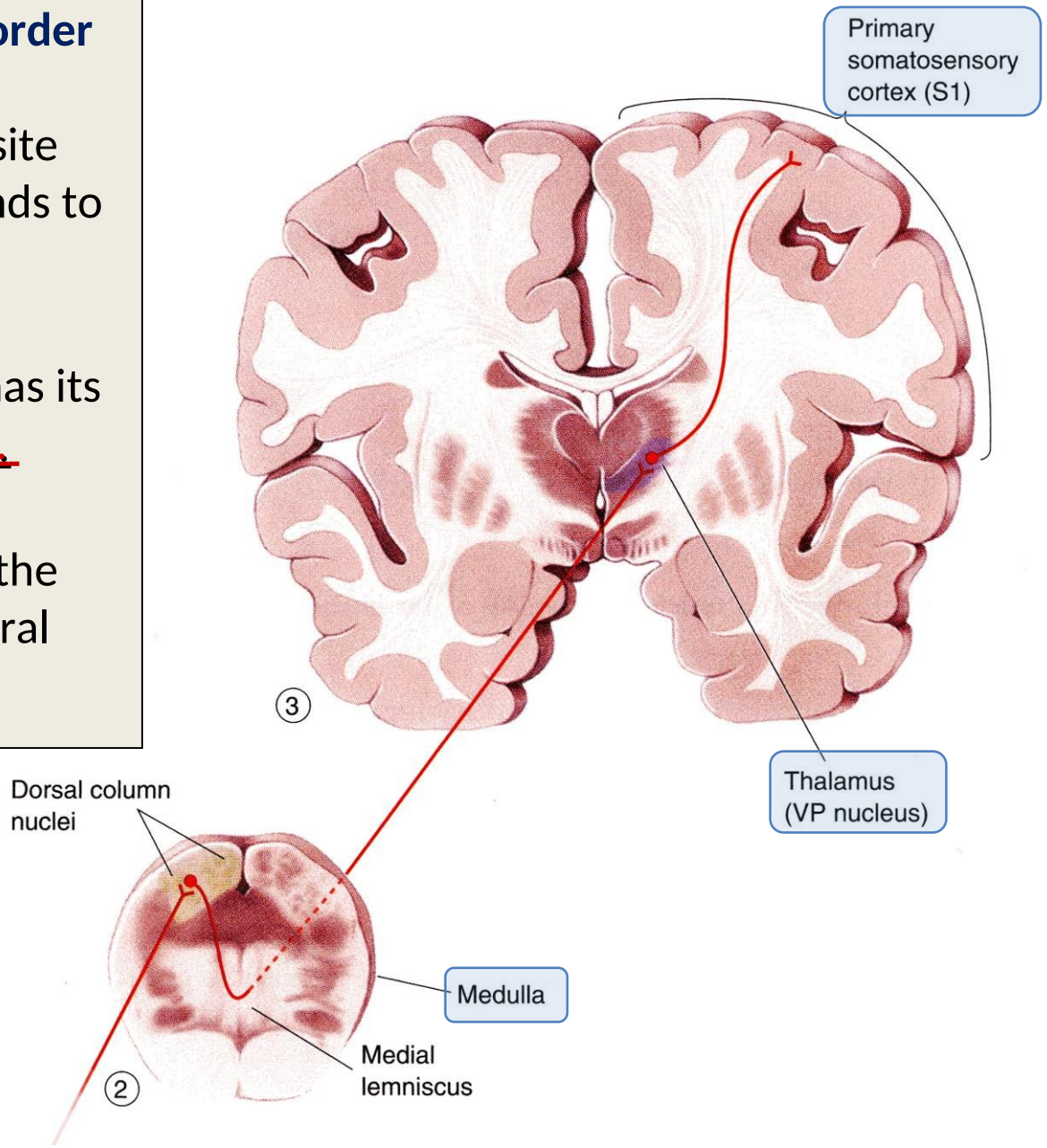

- Pathways that carry information to a **conscious level** share certain common characteristics:
- There is a *sequence of* **Three *Neurons*** between the peripheral receptors and the cerebral cortex.

The **first-order neuron** or primary afferent neuron) enters the spinal cord through the dorsal root of a spinal nerve and its cell body lies in the **dorsal root ganglion.**

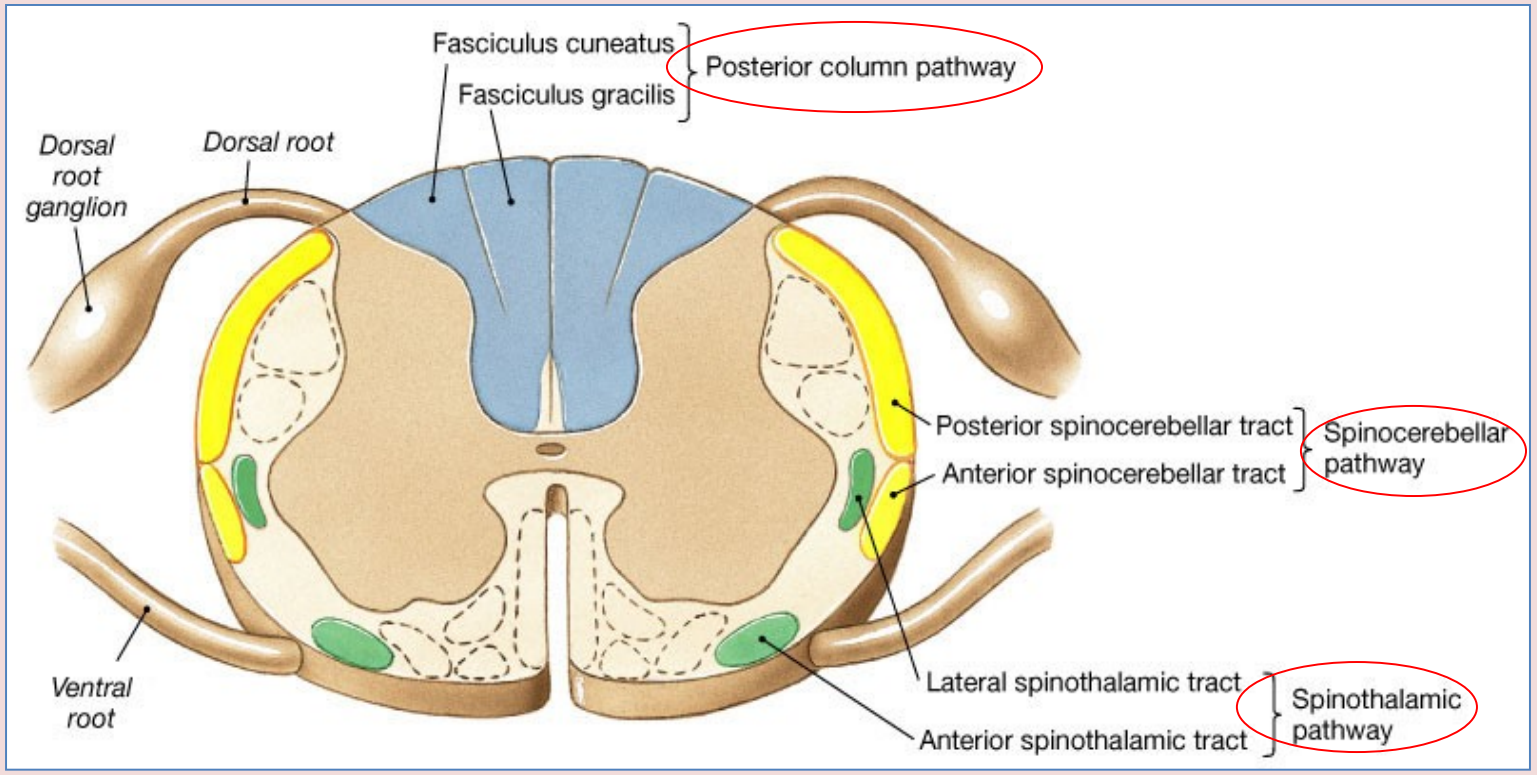
The main fiber remains on the ipsilateral side of the cord and terminates in synaptic contact with the **second order neuron which lies** either in the **spinal grey matter** or in the **medulla oblongata of the brain stem**.



- The axon of the **second order neuron** crosses over (decussates) to the opposite side of the CNS and ascends to the ***thalamus,*** where it terminates.
- **The third-order neuron** has its cell body in the ***thalamus.***
- Its axon passes to the ***somatosensory cortex*** of the parietal lobe of the cerebral hemisphere.

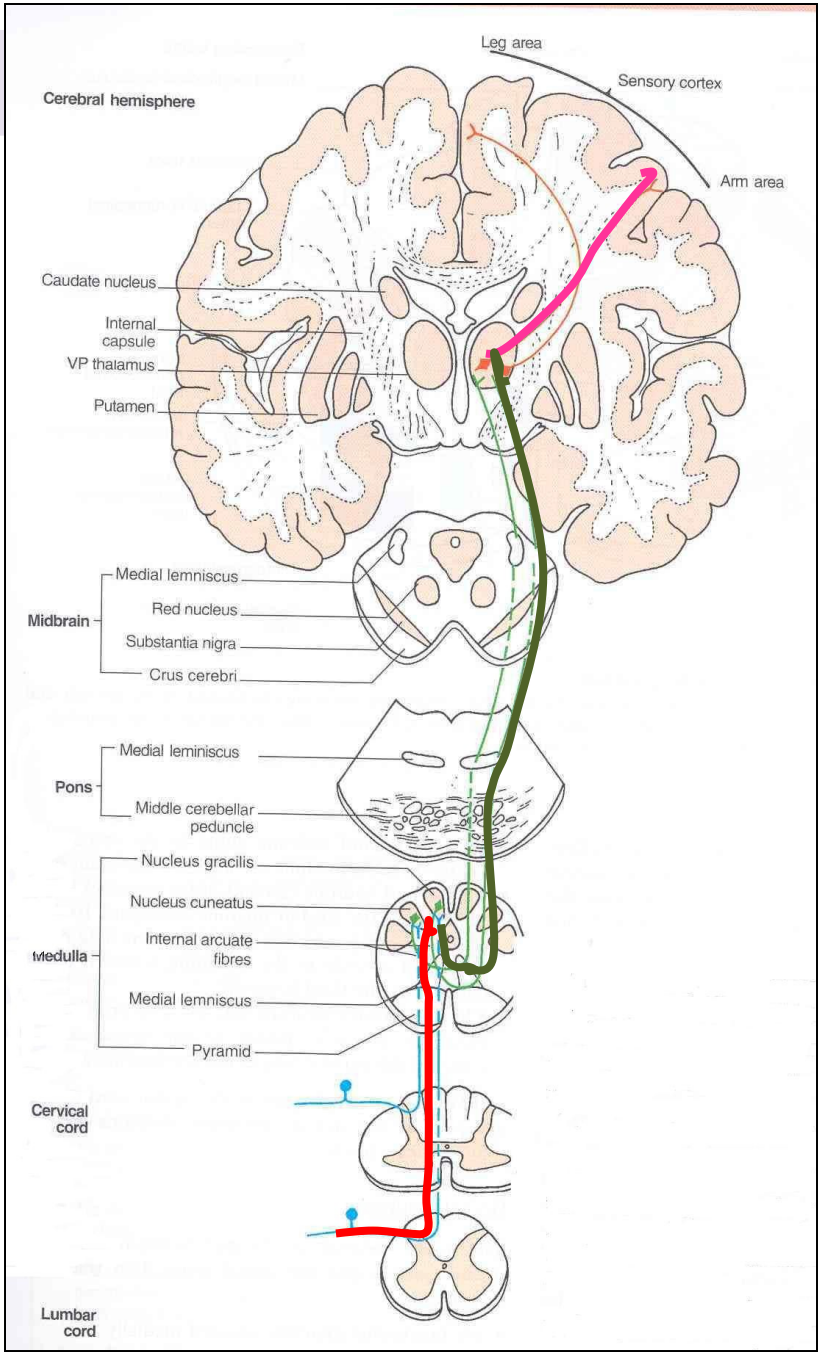


- **Three major pathways** carry sensory information:
  - ***Posterior column*** ***(Gracile & Cuneate fasciculi)***
  - ***Anterolateral pathway (Spinothalamic)***
  - ***Spinocerebellar pathway***

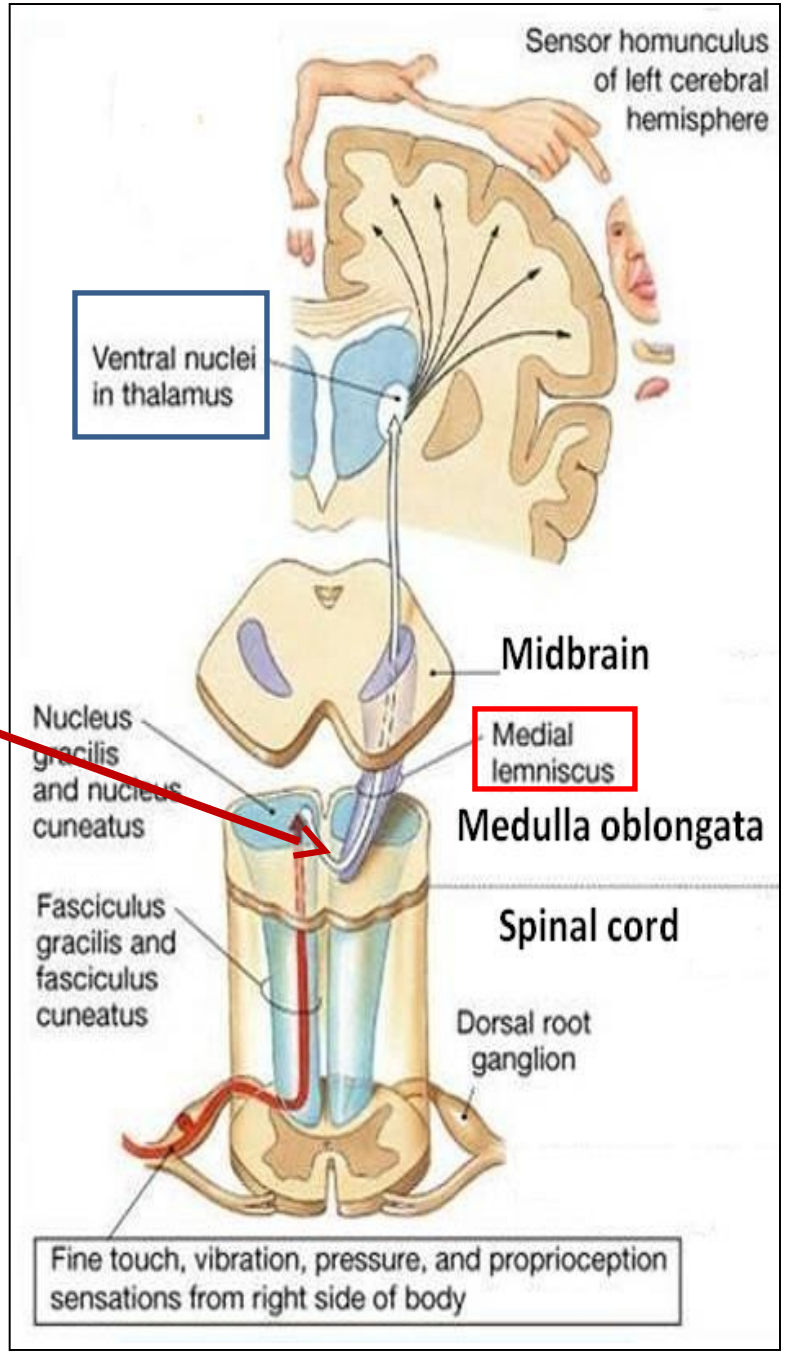


- ***Posterior Column:***
- Contains two tracts; ***Fasciculus Gracilis (FG) & Fasciculus Cuneatus (FC)***
- Carry impulses concerned with **proprioception and discriminative touch.**

- Contain the axons of primary afferent neurons that have entered cord through dorsal roots of spinal nerves
- **FG** contains fibers received from whole length of spinal cord.
- **FC** contains fibers received at upper thoracic and cervical levels

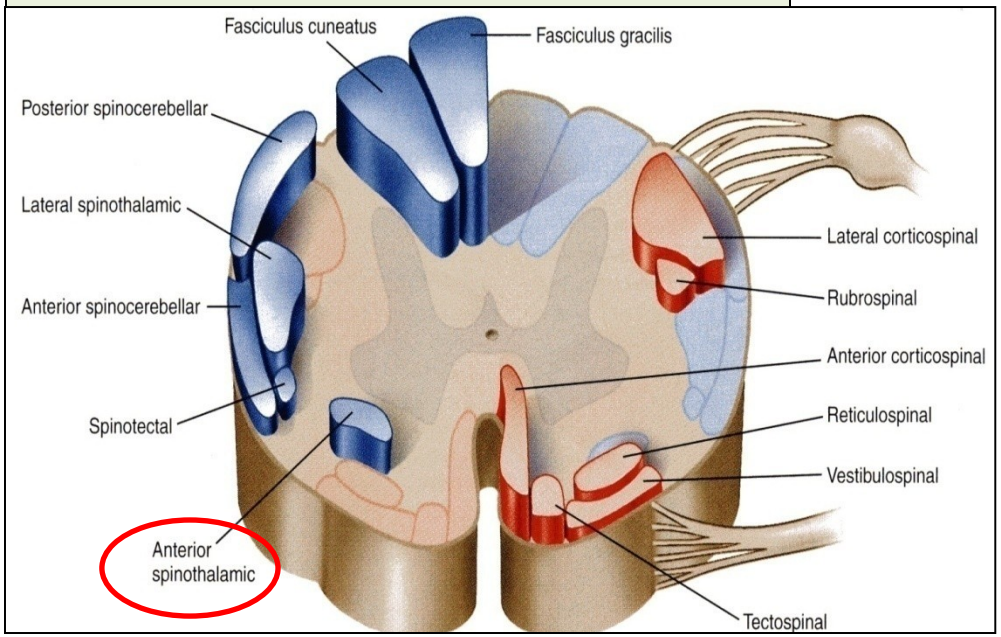


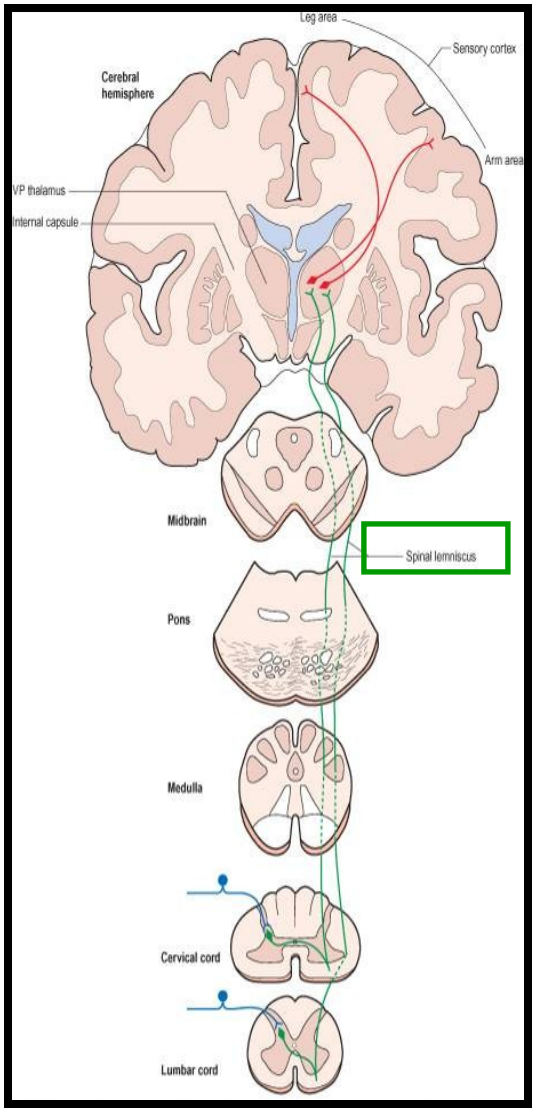
- Fibers ascend without interruption and terminate upon 2nd order neurons in **nucleus gracilis** and **nucleus cuneatus**
- The axons of the 2nd order neurons decussate in the medulla as **internal arcuate fibers** and ascend through the brain stem as **medial lemniscus**.
- The medial lemniscus terminates in the **ventral posterior nucleus of the thalamus (**3rd order neurons), which project to the somatosensory cortex **(thalamocortical fibers)**
-



# Spinothalamic Tracts

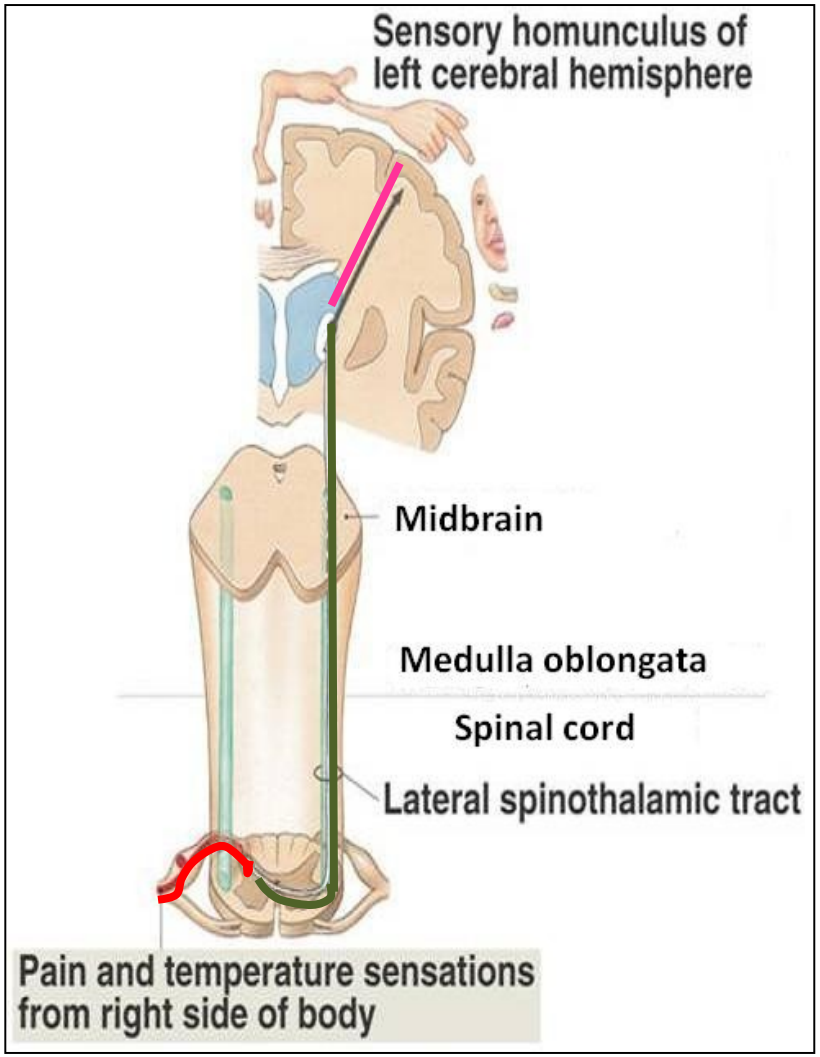
- Located lateral and ventral to the ventral horn.
- Carry impulses concerned with; **pain and thermal sensations (Lateral tract)** and **non- discriminative touch and pressure (Anterior tract).**
- In brain stem, constitute the **spinal lemniscus.**
- Information is sent to the primary sensory cortex on the opposite side of the body



# Lateral Spinothalamic Tract

- **Function:**
- Carries pain & Temperature to thalamus and sensory area of the cerebral cortex.
- **Neurons: 3 Neurons**
- **Neuron I: Small cells** in the dorsal root ganglia.
- **Neuron II:** Cells of **substantia gelatinosa in the posterior horn.**
- **Neuron III:** Cells of **(VP) nucleus of the thalamus.**
- The spinothalamic tract contains ***second-order neurons***, the cell bodies of which lie in the *contralateral* dorsal horn.



# Anterior Spinothalamic Tract

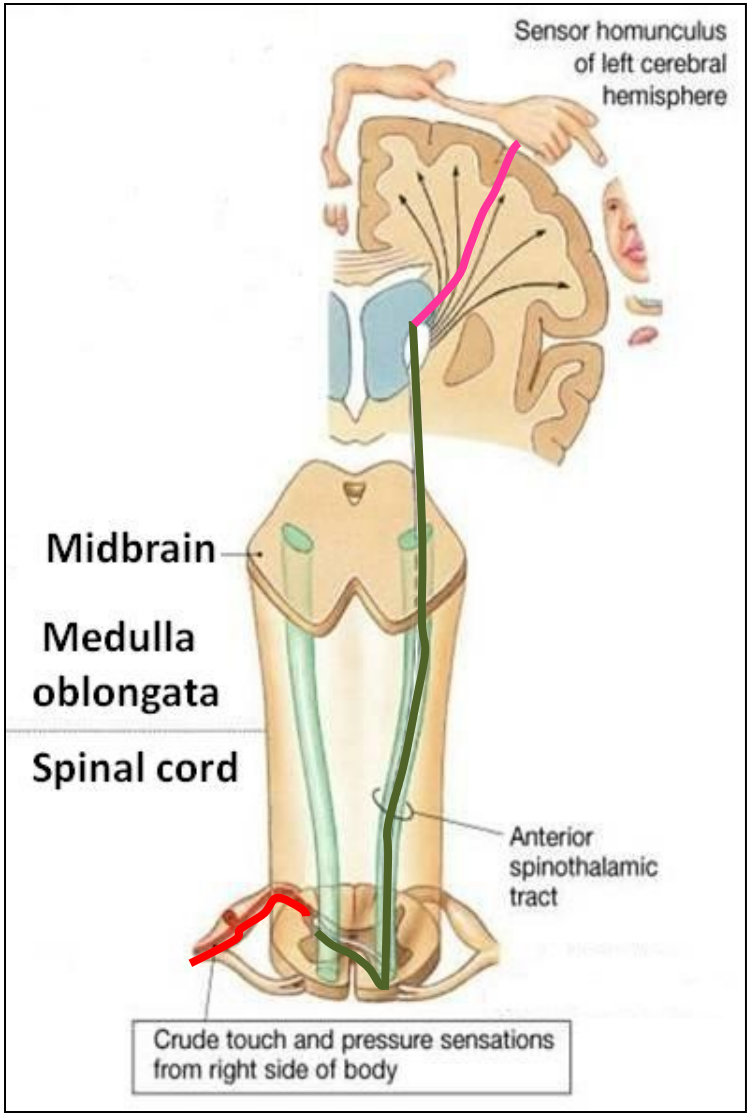
- **Function:**
- Carries crude touch & pressure to thalamus and sensory cortex.
- **Neurones: 3 Neurons**
- **Neuron I:**

  **Medium sized** cells in the dorsal root ganglia.
- **Neuron II:**

  Cells of main sensory nucleus or (**nucleus proprius**).
- **Neuron III:**

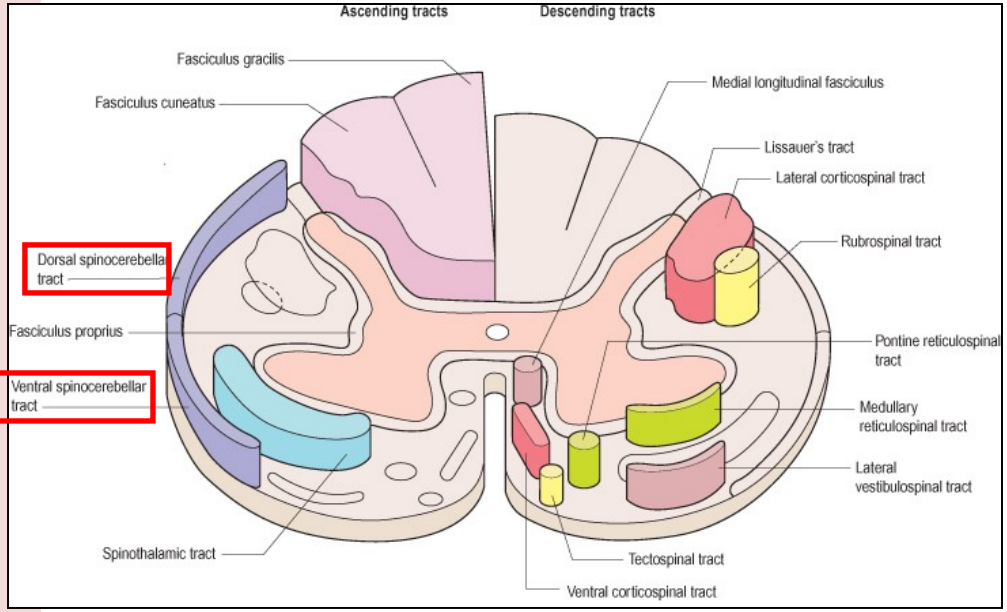
  Cells of VP nucleus of thalamus.
- **Effect of lesion:**

  Loss of crude touch sensation below the level of the lesion.



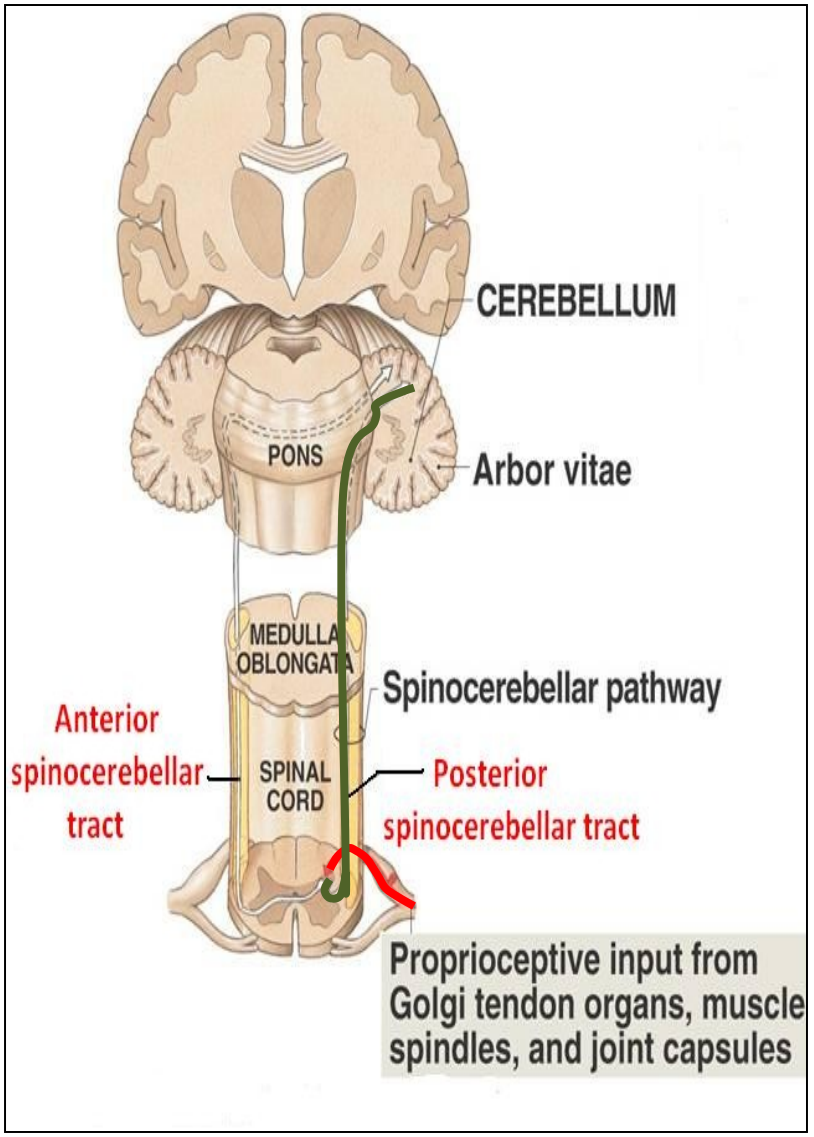
# Spinocerebellar Tracts

- The spinocerebellar system consists of a sequence of **only two neurons;**
- **Neuron I**:Large cells of dorsal root ganglia.
- **Neuron II:** cells of the nucleus dorsalis (Clark's nucleus.
- **Two tracts: Dorsal &Ventral**
- Located near the dorsolateral and ventrolateral surfaces of the cord
- Contain axons of the second order neurons
- Carry information derived from muscle spindles, Golgi tendon organs and tectile receptors to the cerebellum
- for the control of posture and coordination of movements



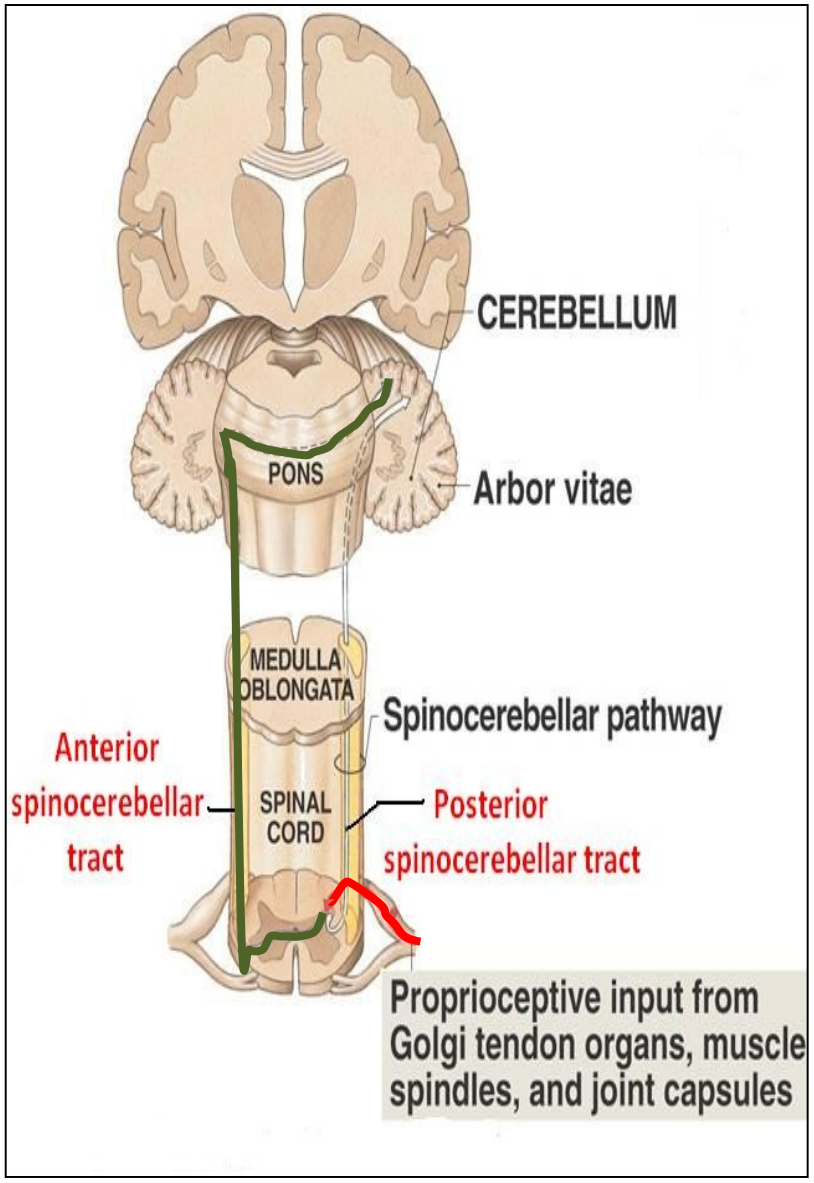
# Posterior Spinocerebellar Tract

- Present only above level L3
- The cell bodies of 2nd order neuron lie in Clark's column
- Axons of 2nd order neuron terminate ipsilaterally (uncrossed) in the cerebellar cortex by entering through the **inferior cerebellar peduncle**.
- **Posterior spinocerebellar tract convey sensory information to the same side of the cerebellum**



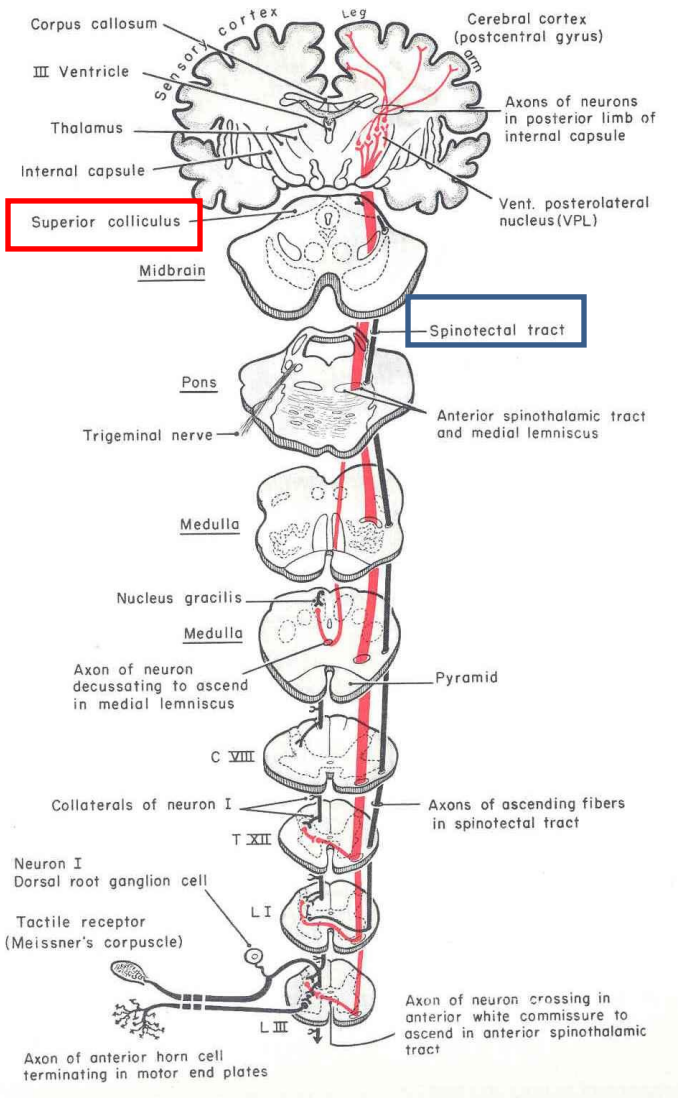
# Ventral (Anterior)Spinocerebellar Tract

- The cell bodies of 2nd order neuron lie in base of the dorsal horn of the lumbosacral segments
- Axons of 2nd order neuron **cross to opposite side,** ascend as far as the midbrain, and then make a sharp turn caudally and enter the **superior cerebellar peduncle**
- The fibers cross the midline for a second time within the cerebellum before terminating in the cerebellar cortex
- Ventral spinocerebellar tract convey sensory information to the same side of the cerebellum



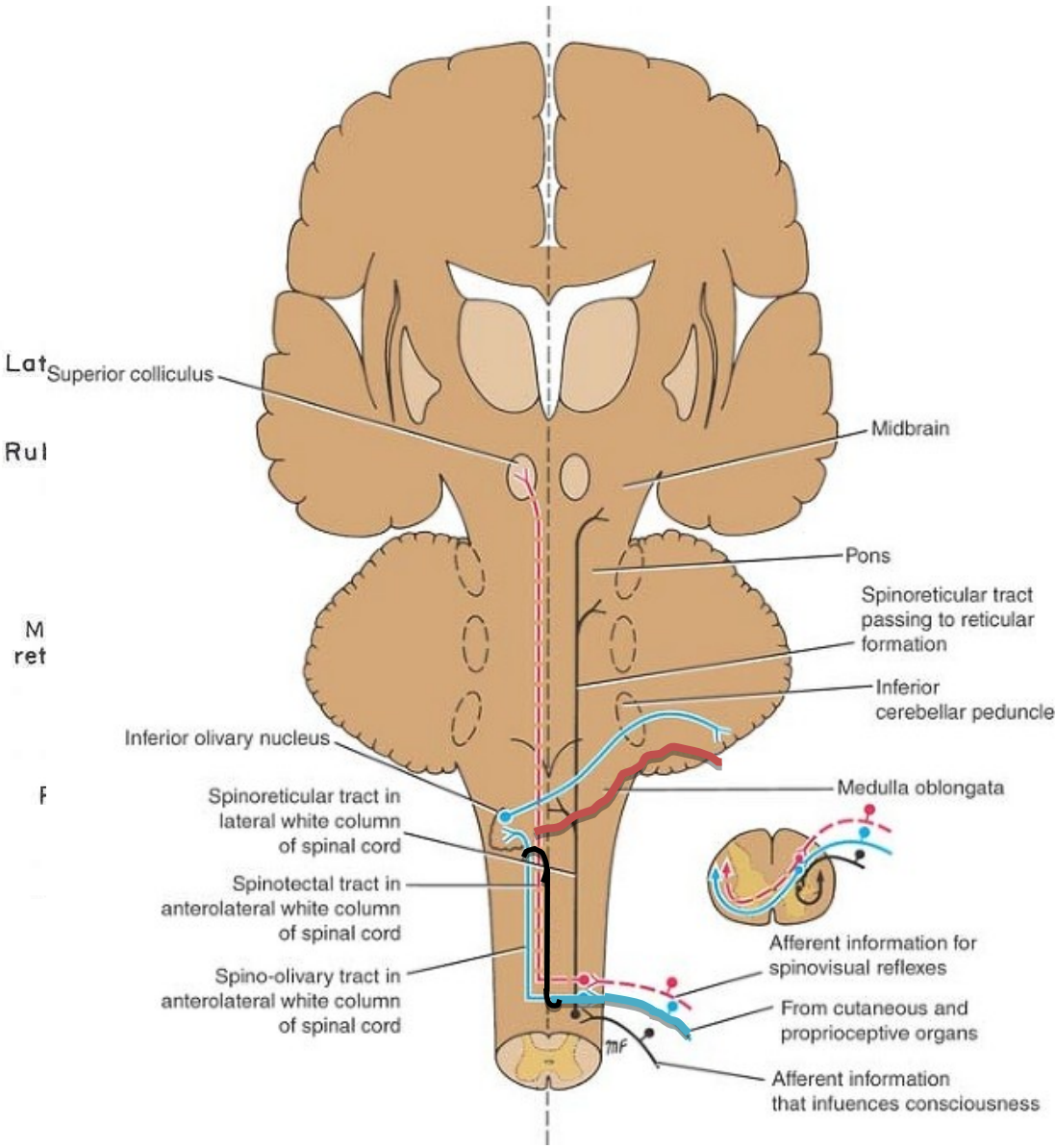
# Spinotectal Tract

- Ascends in the anterolateral part, in close association with spinothalamic system.
- Primary afferents reach dorsal horn through dorsal roots and terminate on 2nd order neurons
- The cell bodies of 2nd order neuron lie in **base of the dorsal horn**.
- Axons of 2nd order neuron cross to opposite side, and project to the periaquiductal gray matter and **superior colliculus in the midbrain**.
- **Involved in reflexive turning of the head and eyes toward a point of cutaneous stimulation.(spinovisual)**



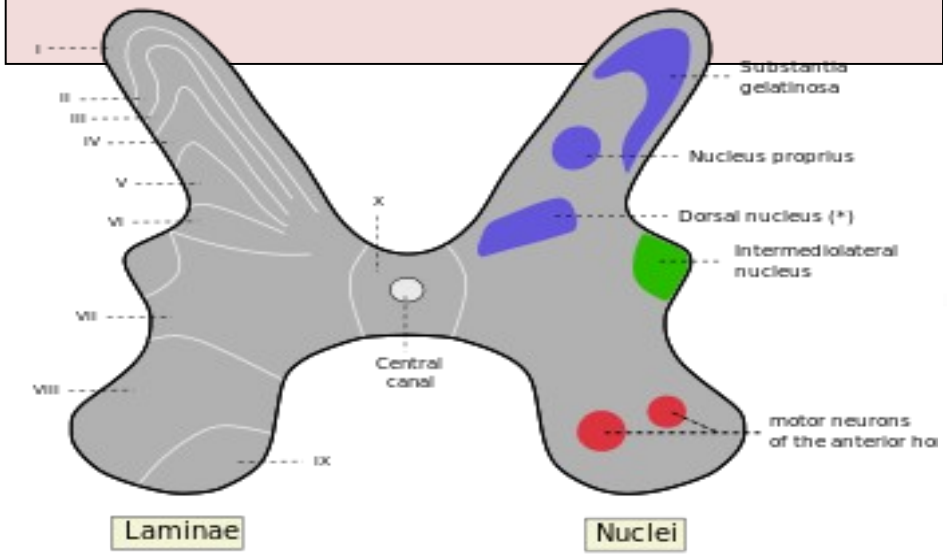
Conveys sensory information to the cerebellum.
Fibers arise at all levels of the spinal cord.
1st order neuron: posterior root ganglion.
2nd order neuron:posterior gray column.axons from here crosses the midline.
3rd order neuron axons synapses in medulla oblongata via inferior olivary nucleus.
Axons of 3rd order cross the midline enters cerebellum via inferior cerebellar peduncle.
**Contribute to movement coordination associated primarily with balance.**

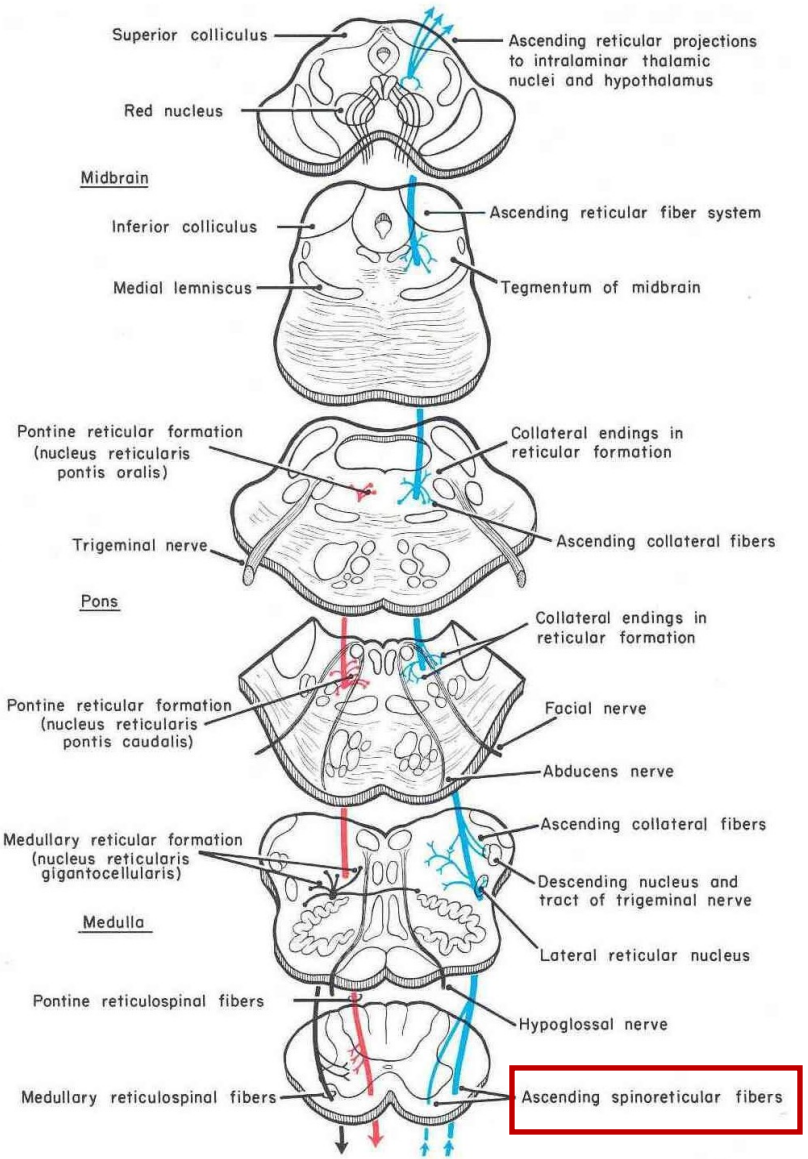
# Spino - olivary Tract



# Spinoreticular Tract

- Originates in laminae IV-VIII
- Contains uncrossed fibers that end in **medullary reticular formation** &
- crossed & uncrossed fibers that terminate in **pontine reticular formation.**
- Forms part of the ascending reticular activating system.
- **Involved in arousing consciousness in the reticular activating system through cutaneous stimulation.**



## Table 15.1 Principal Ascending (Sensory) Tracts and the Sensory Information They Provide

| Tract | Sensations | Location of Neuron Cell Bodies: First-Order | Second-Order | Third-Order | Final Destination | Site of Crossover |
|---|---|---|---|---|---|---|
| **POSTERIOR COLUMNS** | | | | | | |
| Fasciculus gracilis | Proprioception, fine touch, pressure, and vibration from levels inferior to $T_6$ | Dorsal root ganglia of lower body; axons enter CNS in dorsal roots and ascend within fasciculus gracilis | Nucleus gracilis of medulla oblongata: axons cross over before entering medial lemniscus | Ventral posterolateral nucleus of thalamus | Primary sensory cortex on side opposite stimulus | Axons of second-order neurons, before joining medial lemniscus |
| Fasciculus cuneatus | Proprioception, fine touch, pressure, and vibration from levels at or superior to $T_6$ | Dorsal root ganglia of upper body; axons enter CNS in dorsal roots and ascend within fasciculus cuneatus | Nucleus cuneatus of medulla oblongata: axons cross over before entering medial lemniscus | Ventral posterolateral nucleus of thalamus | As above | As above |
| **SPINOTHALAMIC TRACT** | | | | | | |
| Lateral spinothalamic tracts | Pain and temperature sensations | Dorsal root ganglia; axons enter CNS in dorsal roots and enter posterior gray horn | In posterior gray horn: axons enter lateral spinothalamic tract | Ventral posterolateral nucleus of thalamus | Primary sensory cortex on side opposite stimulus | Axons of second-order neurons, at level of entry |
| Anterior spinothalamic tracts | Crude touch and pressure sensations | As above | In posterior gray horn: axons enter anterior spinothalamic tract on opposite side | As above | As above | As above |
| **SPINOCEREBELLAR TRACTS** | | | | | | |
| Posterior spinocerebellar tracts | Proprioception | Dorsal root ganglia; axons enter CNS in dorsal roots | In posterior gray horn: axons enter posterior spinocerebellar tract on same side | Not present | Cerebellar cortex on side of stimulus | None |
| Anterior spinocerebellar tracts | Proprioception | As above | In same spinal segment: axons enter anterior spinocerebellar tract on same or opposite side | Not present | Cerebellar cortex, primarily on side of stimulus | Axons of most second-order neurons cross before entering tract and then cross again within cerebellum |

Thank you